PRE-ALGEBRA (7TH) · UNIT 13: NONLINEAR FUNCTIONS AND POLYNOMIALS

Rabbit Island Population

Model a growing rabbit population with sequences, exponential functions, and polynomials.

CP2

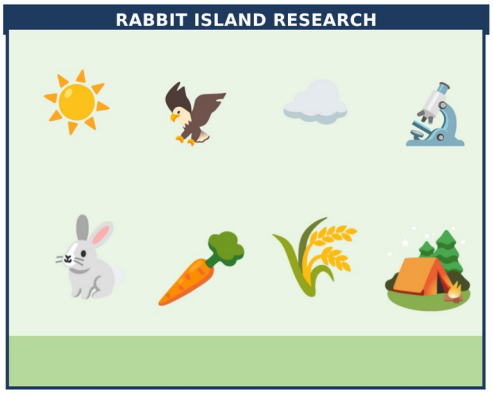
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Ten rabbits were released on a small island with no predators. The population is growing fast, the grass is disappearing, and the wildlife office needs a plan.

You will use sequences, exponential models, and polynomials to predict the population and design habitats.



Start

10 rabbits

Year 0

Growth

Population doubles

every year

Learning Goals

- I can tell arithmetic and geometric sequences apart.
- I can extend a pattern and graph it.
- I can add and subtract polynomials.

Project Timeline

When	What you will do
Day 1	Patterns
Day 2	Graph It
Day 3	Habitat Fences
Final	Wildlife Management Report + reflection

Part 1: Patterns

Day 1 · Use the Population Survey.

Population Survey

Year 0	Year 1	Year 2	Year 3	Year 4
10 rabbits	20 rabbits	40 rabbits	80 rabbits	160 rabbits

1. What do you multiply by each year?

2. Predict the population in years 5 and 6.

3. Grass patches drop by 15 each year: 200, 185, 170, … Is this arithmetic or geometric? What comes next?

4. In which year will the grass patches reach 50?

Part 2: Graph It

Day 2 · Make a table and a graph.

5. Graph the population for years 0–4. Is the graph a line or a curve?

6. Graph the grass patches for years 0–4. Is it a line or a curve?

7. Write a rule for the population after n years.

Starter: *P = 10 × 2^___*

8. Use your rule to find the population in year 8.

Part 3: Habitat Fences

Day 3 · A rectangular pen is x + 4 meters long and x meters wide.

9. Write and simplify the perimeter of the pen.

10. Add the perimeters of two pens: (4x + 8) + (2x + 6).

11. If x = 5, how much fence is needed for both pens?

Final Product: Wildlife Management Report

Your final product must include:

- Population table and graph
- Grass pattern and when it runs low
- Your population rule
- Pen fence expressions
- Report to the wildlife office

Reflection

R1. How is doubling different from adding the same number each year?

R2. Why can the rabbit population not keep doubling forever?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Sequences, exponential models, graphs, and polynomials are correct and used to make a management plan. Goes beyond the minimum with a creative, realistic choice.	Sequences, exponential models, graphs, and polynomials are correct and used to make a management plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes tables, graphs, models, habitat designs, and a recommendation. Polished, organized, and visually engaging.	Report includes tables, graphs, models, habitat designs, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.